Spectral Decomposition of Differential Operators

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Function Decomposition

One result of Fourier Analysis is that the space of square integrable functions $L^2(\mathbb{R})$ has a spectral decomposition. In particular, if $F \in L^2(\mathbb{R}, \mathbb{C})$, then $F = \mathscr{F}^{-1}\{\mathscr{F}\{F\}\} = \mathscr{F}\left\{\mathscr{F}^{-1}\{F\}\right\}$, where $\mathscr{F}$ is the Fourier Transform and $\mathscr{F}^{-1}$ is the Inverse Fourier Transform.

$$\mathscr{F}\{F\}(\omega) = \int_{\mathbb{R}} F(t)e^{-i\omega t}dt \qquad \text{and} \qquad \mathscr{F}^{-1}\{F\}(t) = \frac{1}{2\pi}\int_{\mathbb{R}} F(\omega)e^{i\omega t}d\omega$$

Exponentiation of the Differential Operator

The (partial) differential operator $\partial_t : f \mapsto f_t$, where $f \in C^\infty(\mathbb{R}, \mathbb{C}) \cup C(\mathbb{C})$, will be informally exponentiated. Given that $\tau \in \mathbb{C}$, define $e^{\tau\partial_t}$ by:

$$e^{\tau\partial_t} f := \sum_{n=0}^{\infty} \frac{\tau^n}{n!}\partial_t^n f$$

Because $f \in C^\infty(\mathbb{R}, \mathbb{C}) \subset L^2(\mathbb{R}, \mathbb{C})$, it follows that $f = \mathscr{F}^{-1}\left\{\hat{f}\right\}$, where $\hat{f} = \mathscr{F}\{f\}$. It follows $\forall t \in \mathbb{R}$:

$$e^{\tau\partial_t} f(t) = \sum_{n=0}^{\infty} \frac{\tau^n}{n!}\partial_t^n \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)\sum_{n=0}^{\infty} \frac{\tau^n}{n!}\partial_t^n e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)\sum_{n=0}^{\infty} \frac{(i\omega\tau)^n}{n!}e^{i\omega t}d\omega$$

$$= \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega\tau}e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega(t+\tau)}d\omega = f(t+\tau)$$

It thus follows that $e^{\tau\partial_t}$ is the shift operator $T^\tau : f(t) \mapsto f(t+\tau)$.

Generalized Differential Operators

The generalized differential operator $F(\partial_t)$ is subsequently motivated to be defined $\forall t \in \mathbb{R}$ as such:

$$F(\partial_t)f(t) = \mathscr{F}\left\{\mathscr{F}^{-1}\{F\}\right\}(\partial_t)f(t) := \int_{\mathbb{R}} \mathscr{F}^{-1}\{F\}(\tau)e^{-i\tau\partial_t}d\tau \; f(t) := \int_{\mathbb{R}} \mathscr{F}^{-1}\{F\}(\tau)e^{-i\tau\partial_t}f(t)d\tau$$

$$\implies F(\partial_t)f(t) = \int_{\mathbb{R}} \mathscr{F}^{-1}\{F\}(\tau)f(t-i\tau)d\tau, \; \forall t \in \mathbb{R}$$

Eigen Functions

Let $s \in \mathbb{C}$ and consider applying $F(\partial_t)$ to the exponential function $E_s : t \mapsto e^{st}$, where $t \in \mathbb{R}$:

$$F(\partial_t)e^{st} = \int_{\mathbb{R}} \mathscr{F}^{-1}\{F\}(\tau)e^{s(t-i\tau)}d\tau = \int_{\mathbb{R}} \mathscr{F}^{-1}\{F\}(\tau)e^{-is\tau}d\tau \; e^{st} = \mathscr{F}\left\{\mathscr{F}^{-1}\{F\}\right\}(s)e^{st} = F(s)e^{st}$$

It follows that E_s is an eigen function of $F(\partial_t)$ with eigen value $F(s)$.

Eigen Function Expansion

Because every function f can be expanded as the continuous sum of (a subset of) these eigen functions, in general, applying $F(\partial_t)$ to f equivalently, $\forall t \in \mathbb{R}$, results in:

$$F(\partial_t)f(t) = F(\partial_t)\mathscr{F}^{-1}\left\{\hat{f}\right\}(t) = F(\partial_t)\frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)F(\partial_t)e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)F(i\omega)e^{i\omega t}d\omega$$

Operational Inversion

Suppose there exists an operator $F(\partial_t)^{-1}$ defined on the eigen function(s) E_s, and by extension, all valid functions f, as such:

$$F(\partial_t)^{-1}E_s \coloneqq E_s/F(s) \quad \text{such that} \quad F \neq 0 \quad \text{on} \quad i\mathbb{R}$$

$$\implies \forall t \in \mathbb{R},\; F(\partial_t)^{-1}f(t) = F(\partial_t)^{-1}\frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)F(\partial_t)^{-1}e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\hat{f}(\omega)}{F(i\omega)}e^{i\omega t}d\omega$$

$$\implies \forall t \in \mathbb{R},\; F(\partial_t)^{-1}F(\partial_t)f(t) = F(\partial_t)F(\partial_t)^{-1}f(t) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\hat{f}(\omega)}{F(i\omega)}F(i\omega)e^{i\omega t}d\omega = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(\omega)e^{i\omega t}d\omega = f(t)$$

$$\implies F(\partial_t)F(\partial_t)^{-1} = I = F(\partial_t)^{-1}F(\partial_t)$$

Particular Solutions to Functional Equations

Suppose that $F \neq 0$ on $i\mathbb{R}$ and x is a solution to the functional equation $F(\partial_t)x(t) = f(t),\ \forall t \in \mathbb{R}$. It then follows that $F(\partial_t)^{-1}F(\partial_t)x = F(\partial_t)^{-1}f$, further implying that $x = F(\partial_t)^{-1}f$. Equivalently, $\forall t \in \mathbb{R}$:

$$x(t) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{\hat{f}(\omega)}{F(i\omega)}e^{i\omega t}d\omega$$

Informally, from the theory of differential equations, this is only a particular solution. Furthermore, even if $F = 0$ on $i\mathbb{R}$, so long as the integral has a principal value, x will be a solution. Moreover, as $\mathscr{F}_t\{\delta(t-\tau)\}(\omega) = e^{-i\omega\tau}$, a general formula for the Green's function $G : F(\partial_t)G(t,\tau) = \delta(t-\tau)$ can be obtained as such:

$$G(t,\tau) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{i\omega(t-\tau)}}{F(i\omega)}d\omega$$